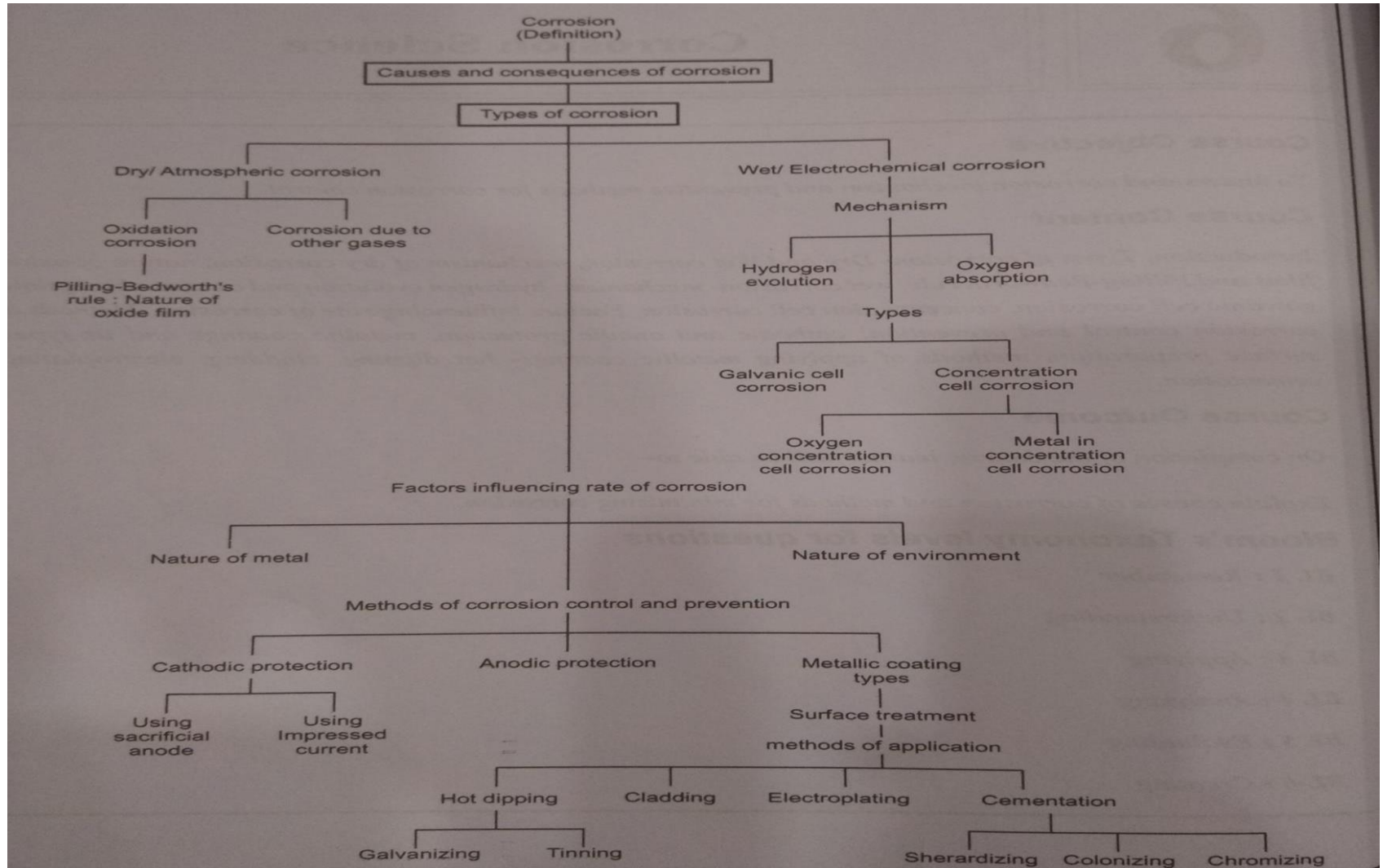
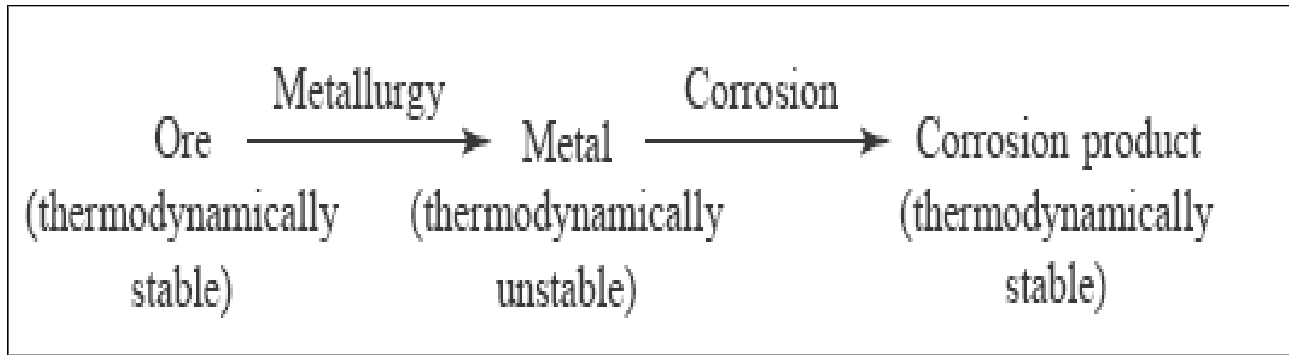


Unit : VI Corrosion Science



1. Definition and causes of corrosion

- **Corrosion may be defined as the destruction of metal by chemical or electrochemical attack of environment, starting at surface**
- **Thermodynamic stability is the only cause behind corrosion process.**
- **Most of the metals occurs in the compound state (except Ag, Au and Pt)**
- **Metal compounds like oxides, chlorides, hydroxide, sulphides, carbonates, silicates etc**



Steel(iron – carbon alloy) are used commonly and on large scale for various metallic designs – no any natural way of protection mechanism from corrosion will be more for prevention and control of steel corrosion.

CORRODED END
Anodic or less noble

Magnesium

Zinc

Aluminum

Cadmium

Steel

Lead

Tin

Nickel

Brass

Bronzes

Copper

Nickel-Copper Alloys

Stainless Steels (passive)

Silver

Gold

Platinum

PROTECTED END
Cathodic or most noble

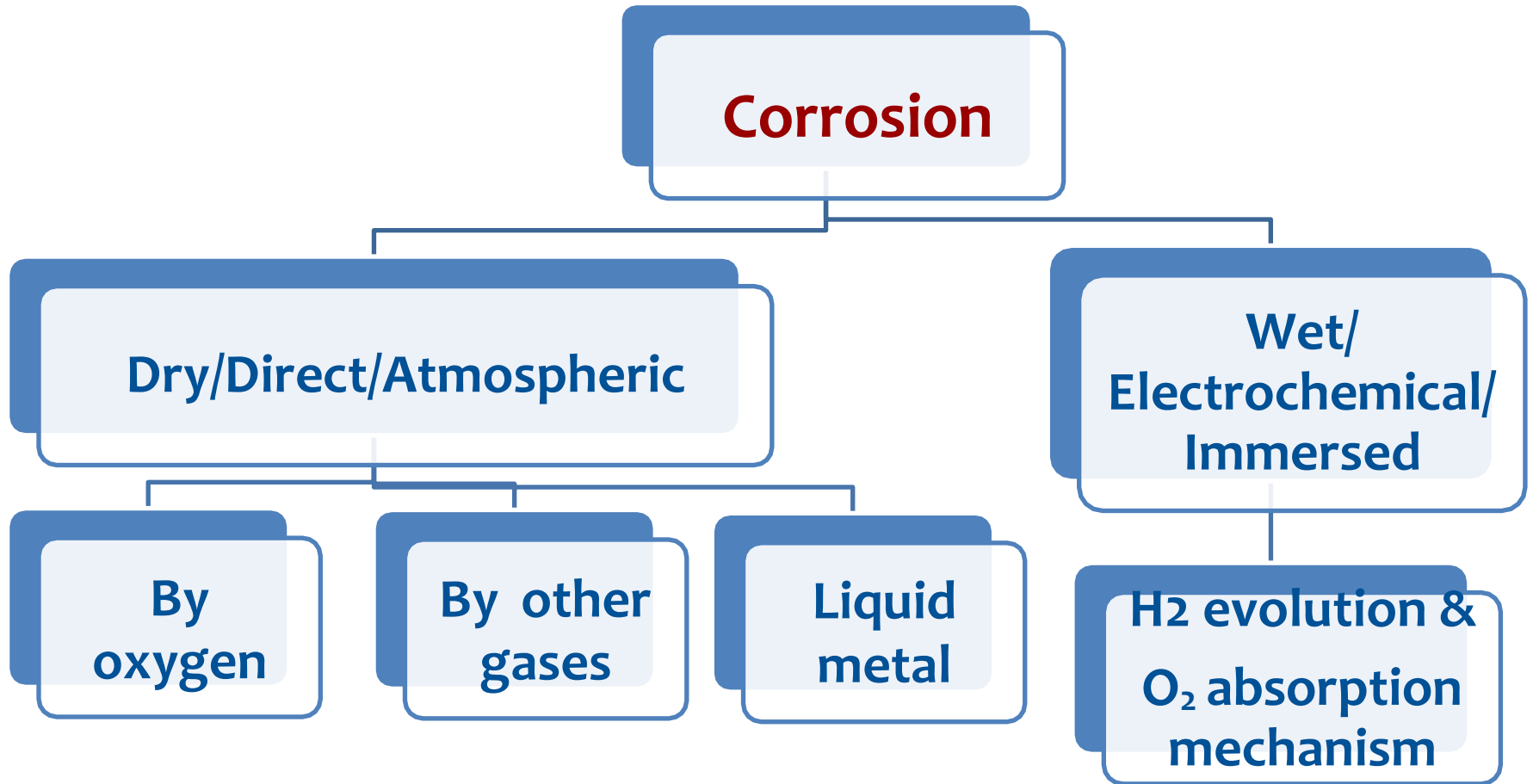
ZINC - Anode

STEEL – Cathode

This arrangement of metals determines what metal will be the anode and cathode when the two are put in an electrolytic cell (arrangement dependent on salt water as electrolyte).

2.

Types of corrosion



1. Dry/Direct/Atmospheric Corrosion

- Due to direct attack of atmospheric gases like oxygen, CO₂, SO₂, H₂S, moisture, ammonia, chlorine, nitrogen oxides etc. and by hot flowing liquids.
- Rate of corrosion depends upon-
- temperature,
- affinity between metal and gas,
- nature of oxide film on surface of metal
- Moisture in the air

H₂S more reactive than O₂ and CO₂

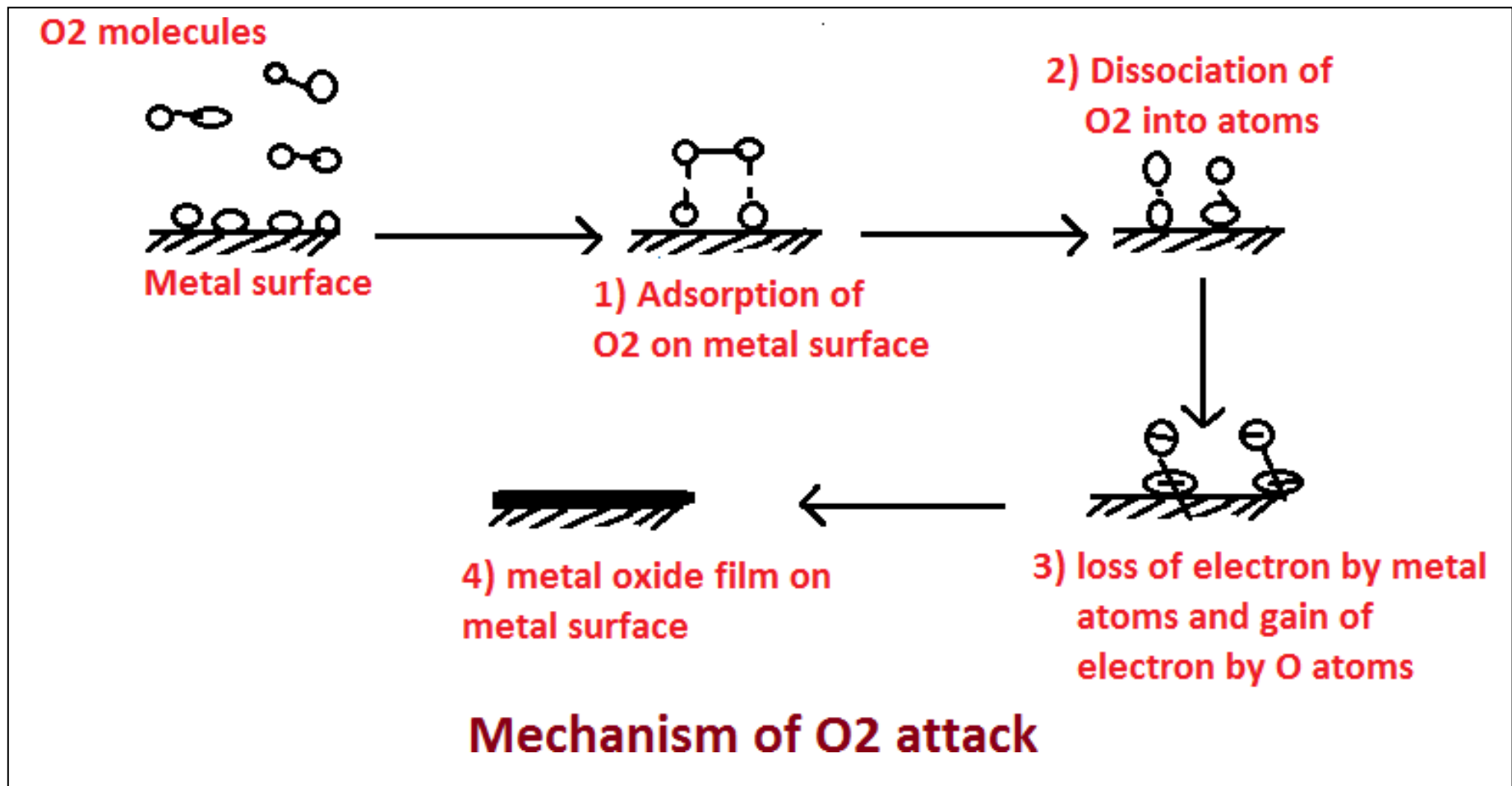
There are three types of dry corrosion

- 1) Corrosion due to oxygen
- 2) Corrosion due to other gases
- 3) Corrosion due to liquid metals

Dry Corrosion

Corrosion due to oxygen (oxidation Corrosion)

- **Mechanism of corrosion by oxygen**
- Oxidation corrosion is brought about by action of oxygen on metal at low or high temperature
- Largest amount of oxygen in air
- The corrosion of metal/ alloy by oxygen takes place in following steps
 - 1) Adsorption of O_2 molecule on metal surface
 - 2) Dissociation of oxygen into its atom
 - 3) Loss of electron by metal atom – results in the formation of metal ion and oxide ion
 - 4) Formation of metal oxide by union of ions



- Thickness of oxide film is less than 300Å

$2M \longrightarrow 2M^{+n} + 2ne^{-}$ Oxidation reaction - loss of electrons

$\frac{n}{2}O_2 + 2ne^{-} \longrightarrow nO^{-2}$ Reduction reaction - gain of electrons

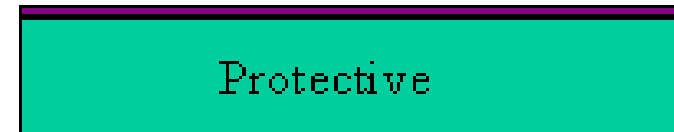
Net reaction - $2M + \frac{n}{2}O_2 \longrightarrow M_2O_n$
Corrosion product,
a metal oxide

- Nature of oxide film i.e. its four types with examples

1. Porous Oxide Film



2. Non Porous Oxide Film



3. Unstable Oxide Film



4. Volatile Oxide Film



Pilling-Bedworth Rule

- Gives idea about nature of oxide film whether it is porous or non-porous.
- Volume of oxide film is smaller than the volume of metal consumed in the metal oxide formation, then the film is porous. And if metal oxide formed has greater volume than the metal consumed during corrosion, then film is non-porous.

$$\text{P.B.R.} = \frac{\text{volume of metal oxide}}{\text{volume of metal consumed}}$$

If $\text{PBR} < 1$ oxide formed is thin and porous- non protective

e.g. Na, Fe, Mg

If $\text{PBR} > 2$ Oxide formed is with cracks – non protective

e.g. Fe

If $1 < \text{PBR} < 2$ oxide formed is nonporous – protective

e.g. Al, Cu, Zn, Cr

Corrosion by other gases

- 1) Halogens
- 2) SO₂, SO₃, H₂S etc
- 3) Corrosion by Hydrogen
- 4) CO and CO₂

Corrosion by liquid metals

1) **Halogen-** First row elements except Fe and Cr are resistant to halogen



Dry chlorine $\rightarrow$ slower action

wet Chlorine $\rightarrow$ enhance the corrosion with the formation of HCl

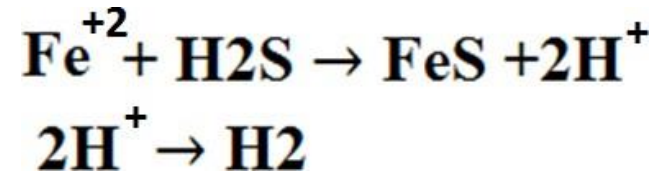
Zirconium shows resistant to dry Cl_2 and under goes corrosion with wet Cl_2

Titanium – resistant to wet Cl_2 and under goes corrosion with dry Cl_2

Ag, Au and Pt - resist the action of halogens at ordinary temperature but attacked at higher temperature

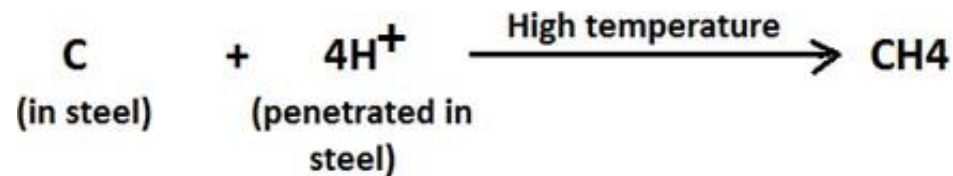
Al and Mg forms protective fluoride film.

2) Corrosion by S – SO₂, SO₃ and H₂S



3) Corrosion By Hydrogen –

Decarburization and hydrogen embrittlement



4) CO and CO₂-



5) By liquid metals-

- i) Dissolution of metal in liquid metal
- ii) Diffusion of liquid metal in solid metal
- iii) Formation of intermetallic compound
- iv) Mass transfer

2.Wet/ Electrochemical Corrosion

- It is also known as indirect or immersion corrosion.
- The corrosion occurring under wet or moist condition through electrochemical reaction i.e. through formation of galvanic cell is known as wet or electrochemical corrosion.
- It can occur within same metal or when two dissimilar metals are in contact with each other.
- **Corrosion of a metal by aqueous conducting medium with the formation of anodic and cathodic areas is known as wet or electrochemical or immersed corrosion**

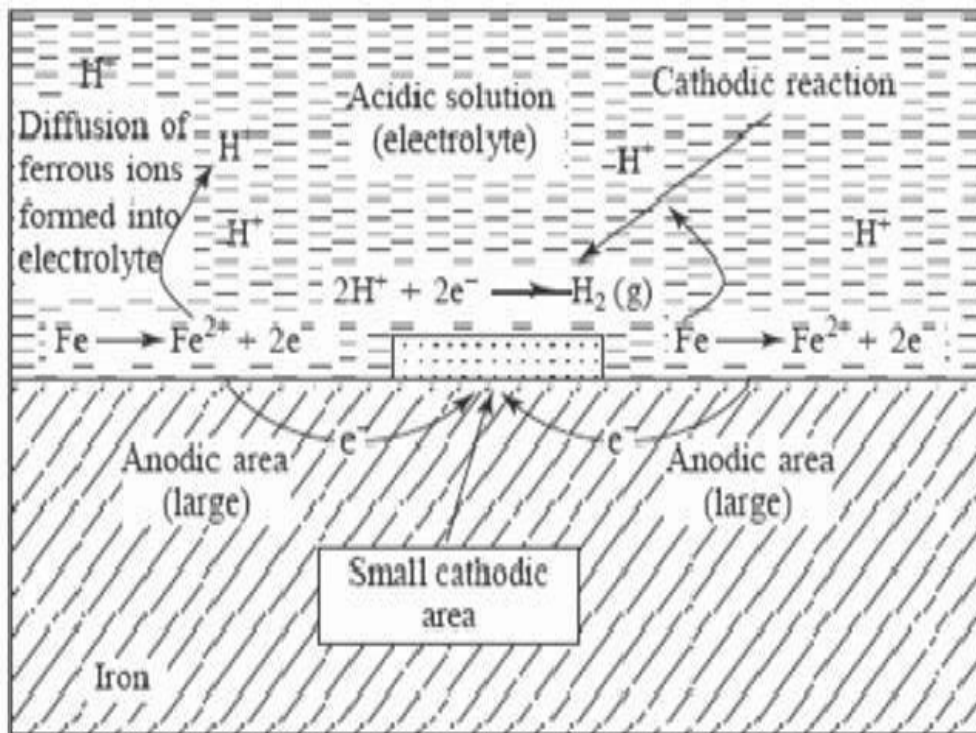
a. Wet corrosion Mechanism by H₂ evolution

Occurs in acidic environment

e.g. steel tank containing acid industrial waste and small copper scrap in presence of acid electrolyte forms electrochemical cell.

Steel – anode (oxidation) of Fe

Copper – cathode (reduction)

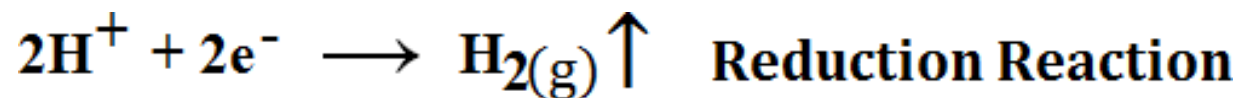


Wet/Electrochemical Corrosion

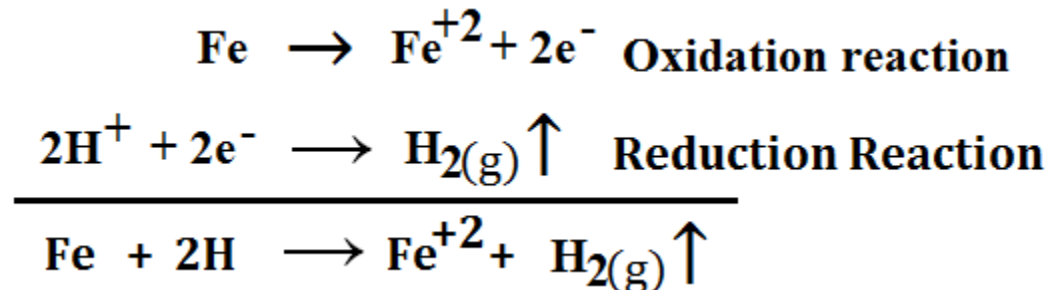
- **Reaction at Anode- Oxidation Reaction**



- **Reaction at Cathode** – In acidic medium (in absence of oxygen) H^{+} ions are formed and at cathode hydrogen ions combine to form hydrogen gas (H_2)



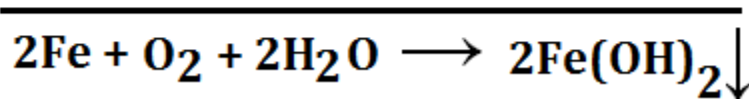
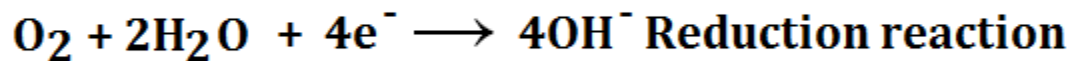
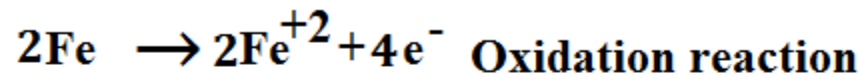
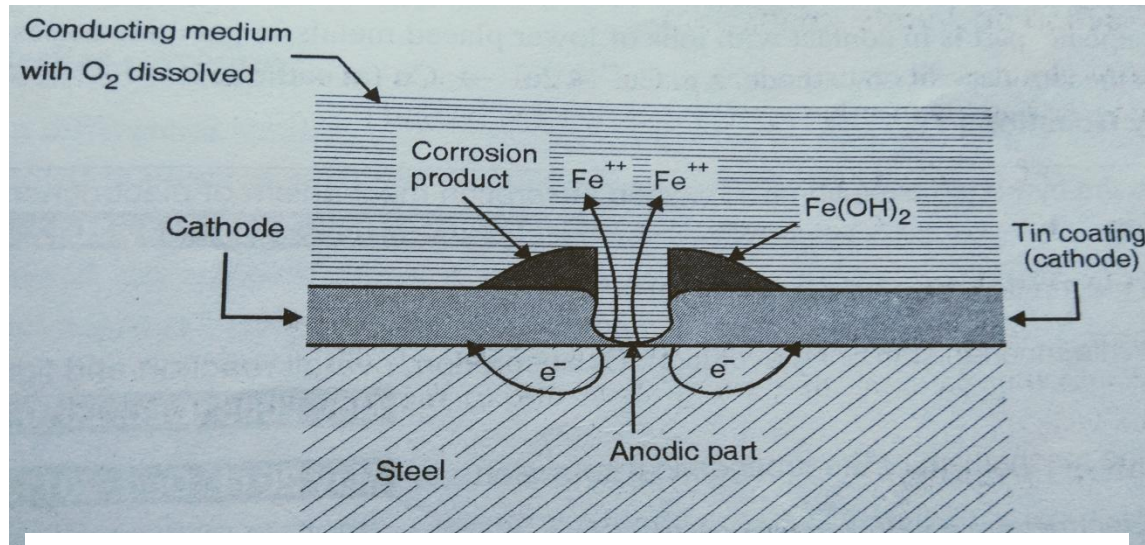
- Overall reaction -



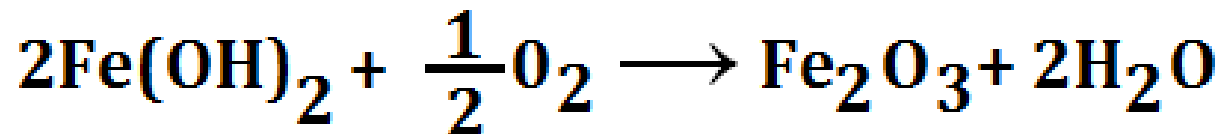
b) Mechanism of O₂ absorption

This type of electrochemical corrosion occurs when electrolyte is neutral aqueous or alkaline solution containing dissolved oxygen.

For example consider a steel plate coated with tin, in due course of time coating at some place get removed, and if there is a neutral or alkaline medium form a galvanic cell.



If enough oxygen is present in medium ferrous hydroxide gets oxidized to Fe_2O_3 rust.



Thus O_2 gas from the aqueous medium is utilized for the reaction and there is formation of metal hydroxide as a corrosion product. If the metal oxide is soluble in aq. medium it remains dissolved and if no soluble then it deposits on cathode area near the anode

- There are two types of wet corrosion

1. Galvanic cell corrosion

2. Concentration corrosion

- 1) Galvanic cell corrosion

1. Two dissimilar metals in contact
2. Stressed and unstressed part of the same metal
3. Impurities in metal.

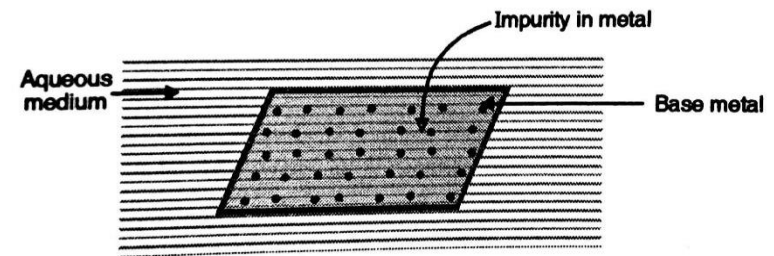
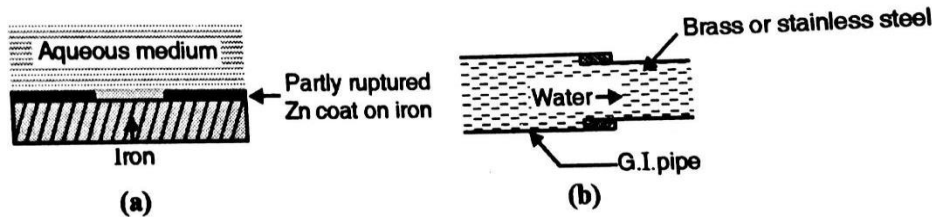


Fig. 6.5.5 : Impure or improperly made alloy (galvanic cells)

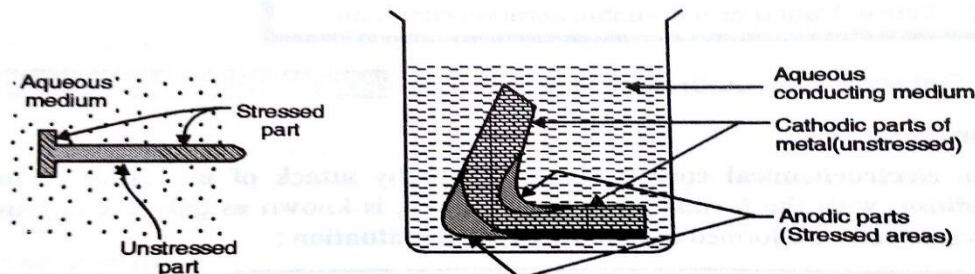


Fig. 6.5.4 : Stressed and unstressed part on a metal (galvanic cell)

2. Concentration cell corrosion

- In all of the following cell less oxygenated part will be acting as anode and get corroded while more oxygenated part will be acting as cathode and get protected.
1. partial immersion of metal
 2. Partial buried in ground
 3. Crevices
 4. Oxide film on the metal surface partially ruptured

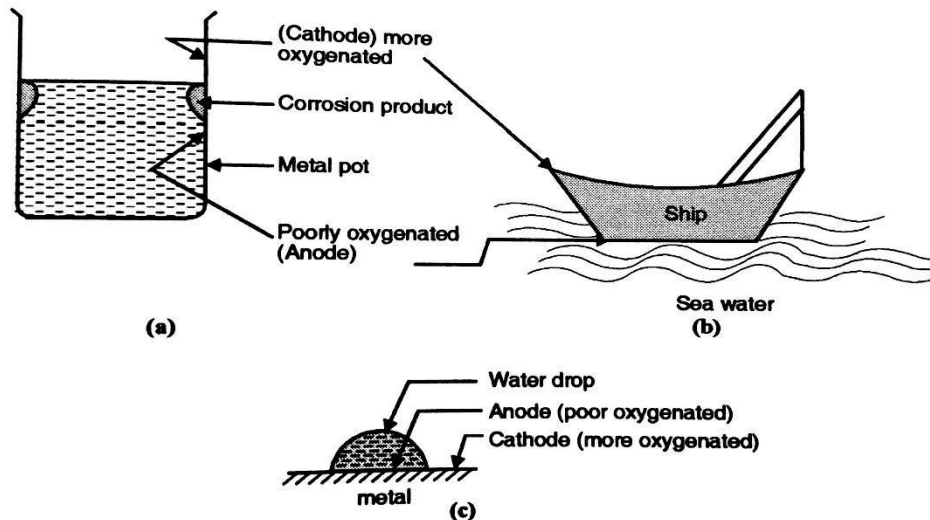


Fig. 6.5.6 : Partial immersion in water (Diff. aeration cells)

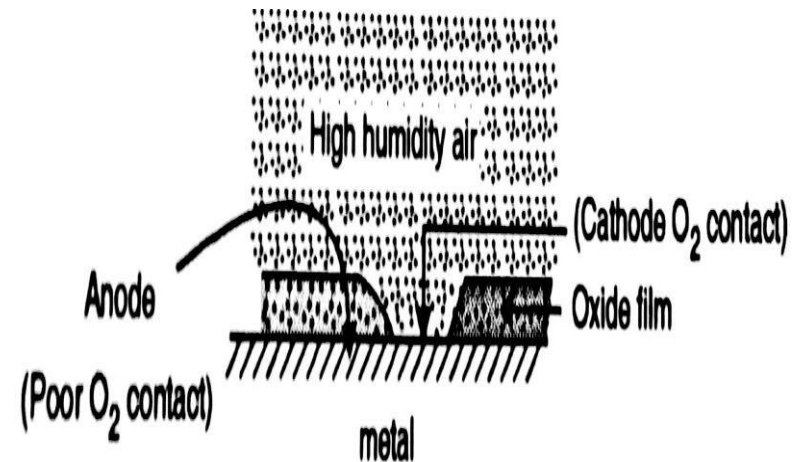


Fig. 6.5.7

Electrochemical and Galvanic Series

Electrochemical Series

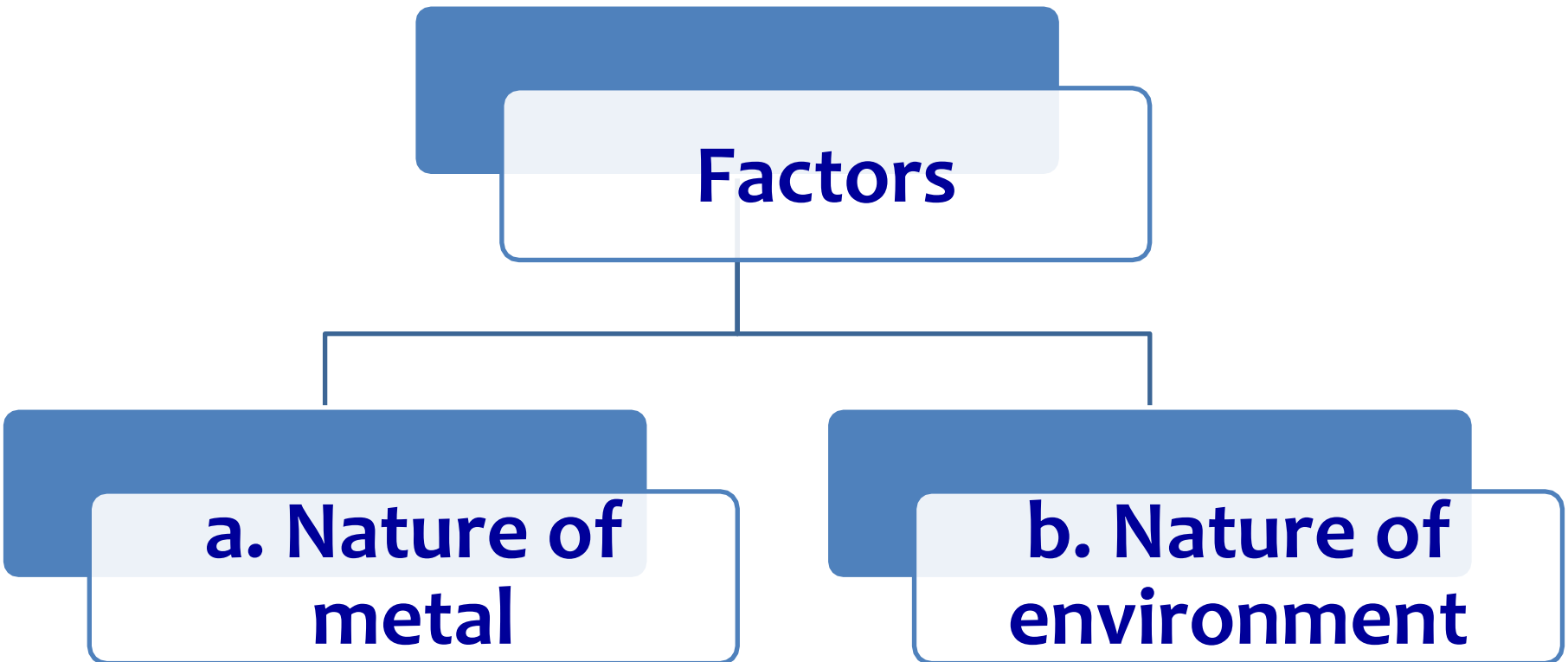
1. It predicts the relative displacement tendencies
2. Standard hydrogen electrode is used as reference electrode
3. Position of metal is fixed. That cannot be changed
4. concentration of salts of the same metal as electrolyte.
5. Electrode potentials measured only for metals and nonmetals

Galvanic Series

1. It predicts the corrosive tendencies of metal alloys.
2. Calomel electrode is used as a reference electrode
3. Positioning of metal or alloy may change
4. The metals and alloys are immersed in the sea water for study.
5. Electrode potentials are measured for both metals and alloys.

3.

Factor affecting rate of corrosion



a. Nature of metals

1. Position of metal in galvanic series: It decides the corrosion rate. A metal having higher position in galvanic series undergoes corrosion when connected to another metal below it. Also, more difference in the position of galvanic series will cause faster corrosion at anodic metal.

2. Hydrogen over voltage:

In acidic medium wet corrosion occurs by H_2 evolution mechanism in which anode gets dissolved and H_2 gas is liberated at cathode.

At this potential hydrogen gas is in equilibrium with hydrogen ions. This potential can be theoretically calculated. But experimentally it is observed that actual evolution of H_2 gas at cathode does not occur at this calculated potential but requires higher potential.

This is because dissolving of anode occurs at the theoretically calculated potential, formation of hydrogen gas from hydrogen ions is slow and requires some additional voltage called activation potential. As a result decomposition potential becomes high.

The difference between the theoretical decomposition potential and the actual potential at which hydrogen gas is liberated at cathode is known as hydrogen over voltage.

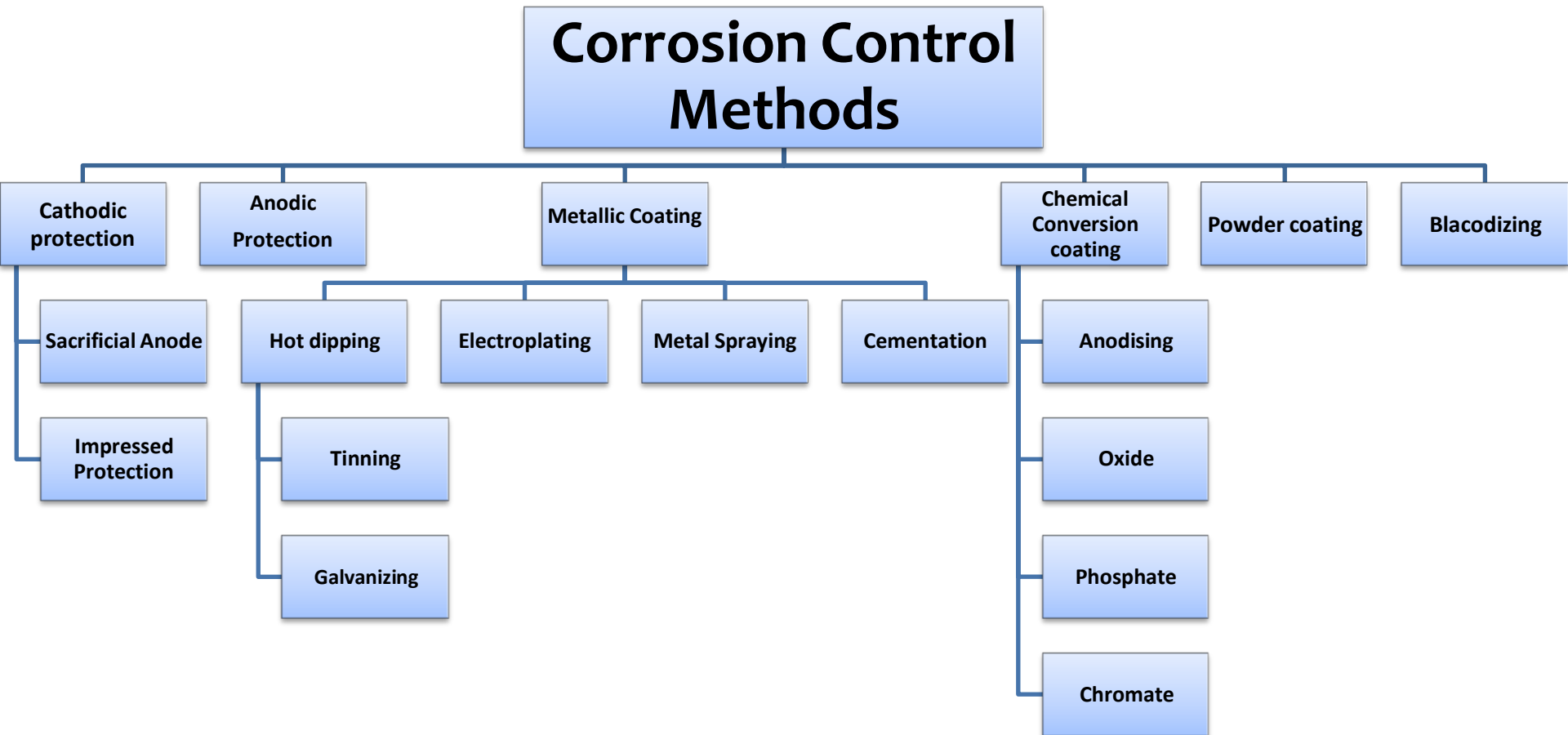
The value of over voltage depends upon the nature of metal, nature of electrolyte and current density. Higher is the over voltage lesser is the rate of corrosion.

3. **Purity of metal:** Pure metal resists corrosion, while impurities in a metal form a local galvanic cell (metal as anode and impurity as cathode) and result in the corrosion of metal. Traces of impurities causes a very rapid corrosion. For alloys the system is a homogeneous solid solution, hence no local action and no corrosion.
4. **Relative areas of anode and cathode:** Smaller the area of anode compared to cathode will lead to faster corrosion of anode.
5. **Physical state of the metal:** Small granular metal will corrode faster than the larger one. Also the type of structure formed by a metal will have effect on the corrosion rate. A bent metal (stress) is rapidly corroded due to stress.
6. **Nature of oxide film:** An oxide film is formed by the reaction between metal and oxygen. If this oxide film is porous and oxygen can be diffused through it, more corrosion is expected (already shown in dry or chemical corrosion). Also if volume of metal oxide is more than the volume of metal (The specific volume ratios of Ni, Cr, W are 1.6, 2.0 and 3.6) least corrosion or no further corrosion occurs.
7. **Nature of corrosion product:** In electrochemical corrosion if the corrosion product is soluble in corroding medium, corrosion proceeds at a faster rate. If corrosion product is insoluble, forms a barrier and stops the further corrosion. If corrosion product is volatile, corrosion continued.

b. Nature of environment

1. **Temperature-** Temperature increase rate of corrosion increases.
2. **Humidity-** Corrosion is fast in humid atmosphere .
3. **pH value-** comparatively in acidic medium rate of corrosion is fast.
4. **Nature of ions** if Cl^- is present in the solution it will destroy the protective coating and exposing the surface for corrosion. And silicate is present then it will act as a barrier for corrosion
5. **Conductance of medium.** clayey soil – more corrosive and sandy soil – less corrosive. Rate of corrosion is high in highly conducting medium
6. **Formation of oxygen concentration cell.**

Methods of Corrosion Control



A. Cathodic Protection:

Principle - In electrochemical corrosion, anode is the one which undergoes corrosion and cathode remains protected from corrosion in cathodic protection, **Metal to be protected is forced to behave as cathode.**

Two ways -

(i) Sacrificial anodic protection or galvanic protection :

- Metal to be protected from corrosion connected to more anodic metal
- Commonly used metals Mg, Zn, Al and their alloys

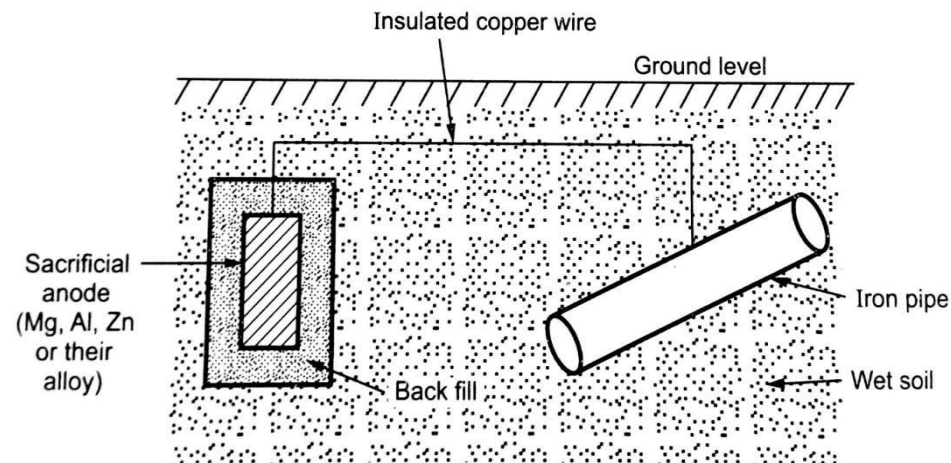


Fig. 6.10.1 Sacrificial anodic protection of underground pipeline

Advantages

1. No external power is required.
2. Easy to install.
3. Anodes can be readily added .
4. Minimum maintenance required.

Limitations

1. Limited driving potential.
2. Limited current output.
3. It can be ineffective in high resistivity environment.
4. Poorly coated structure may require many anodes

Applications-

1. buried steel pipeline
2. Ship Hull
3. hot water tanks
4. buried cables

(ii) Impressed current method:

- Direct current is applied in opposite direction to nullify the corrosion current
- Converts the corroding metal from anode to cathode.
- Negative terminal of the DC is connected to pipeline to be protected
- **Applications** – 1. open water box coolers 2. water tanks 3. buried 4. condensers 5. transmission line towers

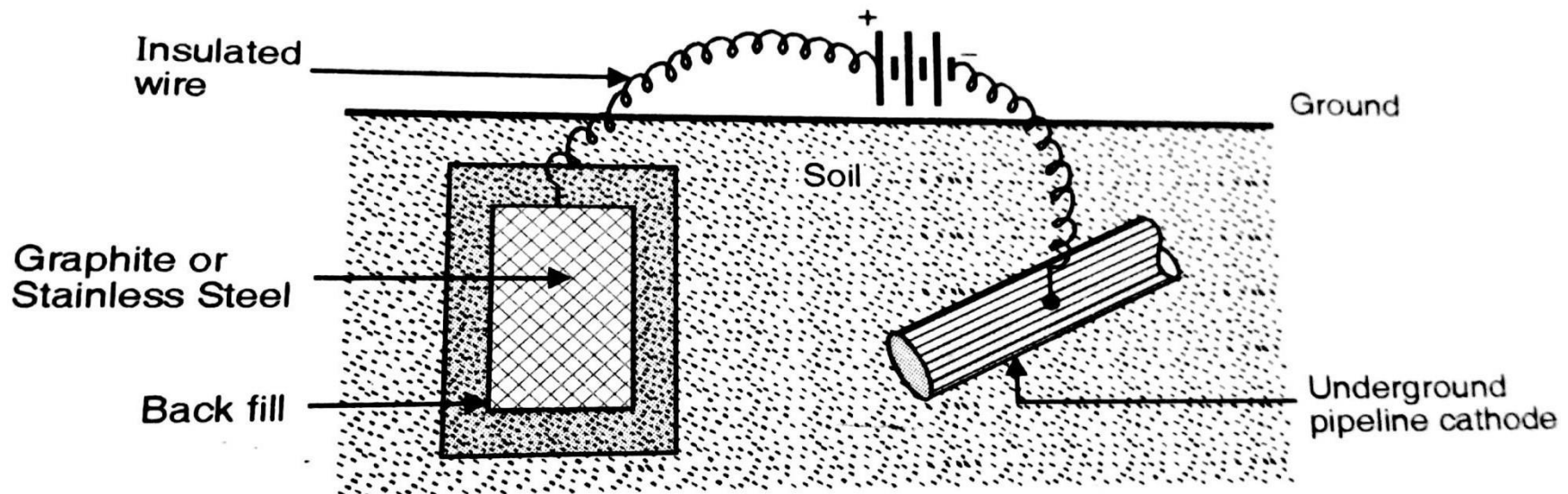


Fig. 6.9.2 : Cathodic protection by impressed current

- Impressed current method is particularly useful where current requirement and resistivity are high. This is suited to large structures and long term operations

Advantages

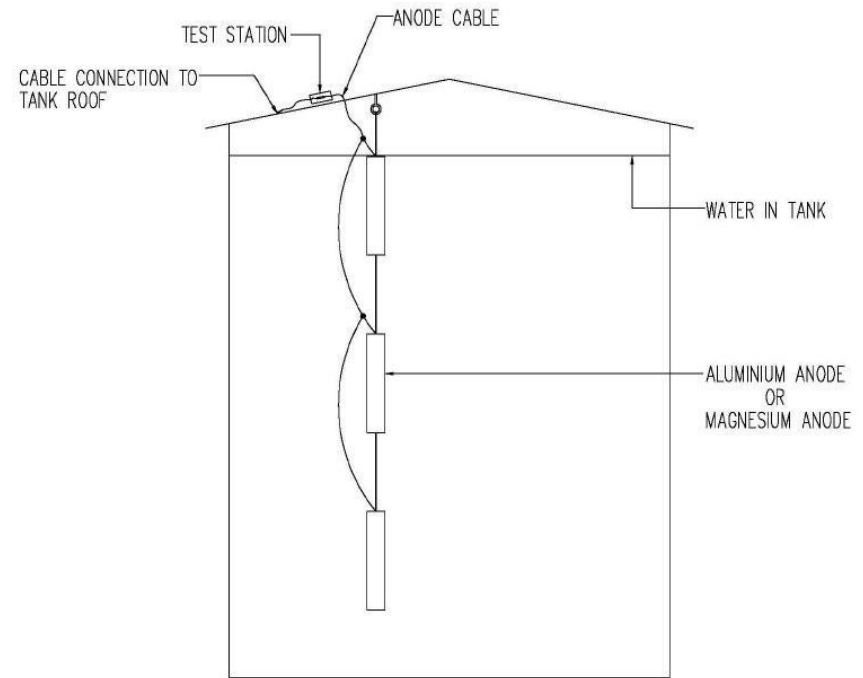
1. Large structures can be protected for long term operations.
2. Can be designed for wide range of current and voltage.
3. It can be applied in high resistivity environment.
4. Effective in protecting uncoated and poorly coated environment.

Limitations

1. It can cause cathodic interference problems.
2. Require periodic maintenance.
3. Require external power, resulting in monthly power costs.
4. Over protection can cause coating damage.

- Applications-

- 1. buried steel pipeline
- 2. Ship Hull
- 3. hot water tanks
- 4. buried cables



TANK INTERNAL
SACRIFICIAL STRING ANODE CATHODIC PROTECTION SYSTEM



2. Anodic protection

Principal -in this method the metal to be protected make it more anodic by passing D.C. in the direction of corrosion current as a result oxide film is formed which protects the metal from further corrosion. Passivity of the metal is quite common for Al, Cr, Ni, Ti, etc

Working – the potentiostat is adjusted for particular current and potential so that anodic metal tank gets oxidized and passivated. There is a formation of thin corrosion resistant oxygen layer which protects the metal from corrosion. once the metal become passive potential changes to high positive value and current reduces. Very small current is required to maintain passivity of the metal.

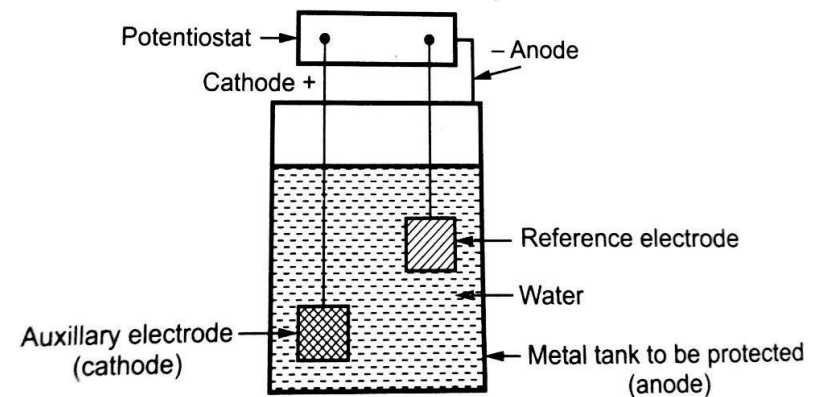


Fig. 6.11.1 Anodic protection of water tank

- Advantages

1. Method can be applied to severely corroding medium like conc. Acid
2. It requires low current density, hence its operation cost is low

- Limitations

1. Applicable to only those metals which exhibit passivity.
2. During anodic protection take place in small amount but not zero.

- Applications

1. Chemical reactors, tanks
2. Pipes carrying corrosive liquids.
3. Pipes carrying concentrated acids.

Comparison of cathodic and anodiic protection

Sr. No.	Cathodic protection	Anodic protection
1.	Method applicable to all metals.	Method applicable to only those metals which show active - passive behaviour.
2.	Used for metal protection in less electrical conductance of environment.	Useful for metal protection in highly conducting environments.
3.	Installation cost low.	Installation cost higher.
4.	Operating cost higher.	Operating cost lower.
5.	If system goes out of control, slower corrosion of metal.	If system goes out of control quite fast corrosion of metal.
6.	Feasibility cannot be predicted by laboratory experiments.	Feasibility and design can be predicted by laboratory experiments.

Metallic Coating

- Surface treatment before metallic coating
 - 1) **Solvent cleaning** - to remove oil/ greasy matter by use of organic solvent or detergents
 - 2) **Alkali cleaning** – to remove old paint, rust
 - 3) Then **Acid cleaning** and finally **water washing** and drying.
 - 4) **Flame cleaning**- to remove loose scale and moisture followed by wire brushing, sand blasting

Metallic coatings are of two types

- i) Anodic Coating
- ii) Cathodic coating

1) Anodic coating – when coating metal is higher placed in galvanic series than base metal, then coating is called as anodic coating.

e.g. Zn, Al, Cr coated on steel

2) Cathodic coating - when coating metal is lower placed in galvanic series than base metal, then coating is called as cathodic coating.

e.g. coating of tin on steel, silver or gold on brass etc .

Anodic coating is most preferable than cathodic coating.

Anodic Coating	Cathodic coating
1) Coating metal is in higher placed in EMF series than base metal	1) Coating metal is in lower placed in EMF series than base metal
2) Base metal remain protected even if coating metal get rupture and if conducting medium is present on base metal	2) Base metal remain safe/protected as long as coating metal is present on base metal , ones coating get rupture and if aqueous conducting medium is there base metal starts corroding .
3) Anodic coating most preferred.	3) Cathodic coating less preferred.

Metallic Coating

Metal can be protected by some protective coating material. Coating can be metallic or non- metallic.

Methods of Application of Metallic Coating

- 1) Hot dipping
- 2) Metal Spraying
- 3) Electroplating
- 4) Cementation

1) Hot dipping – Coating Metal is kept in molten state and base metal is dipped into it.

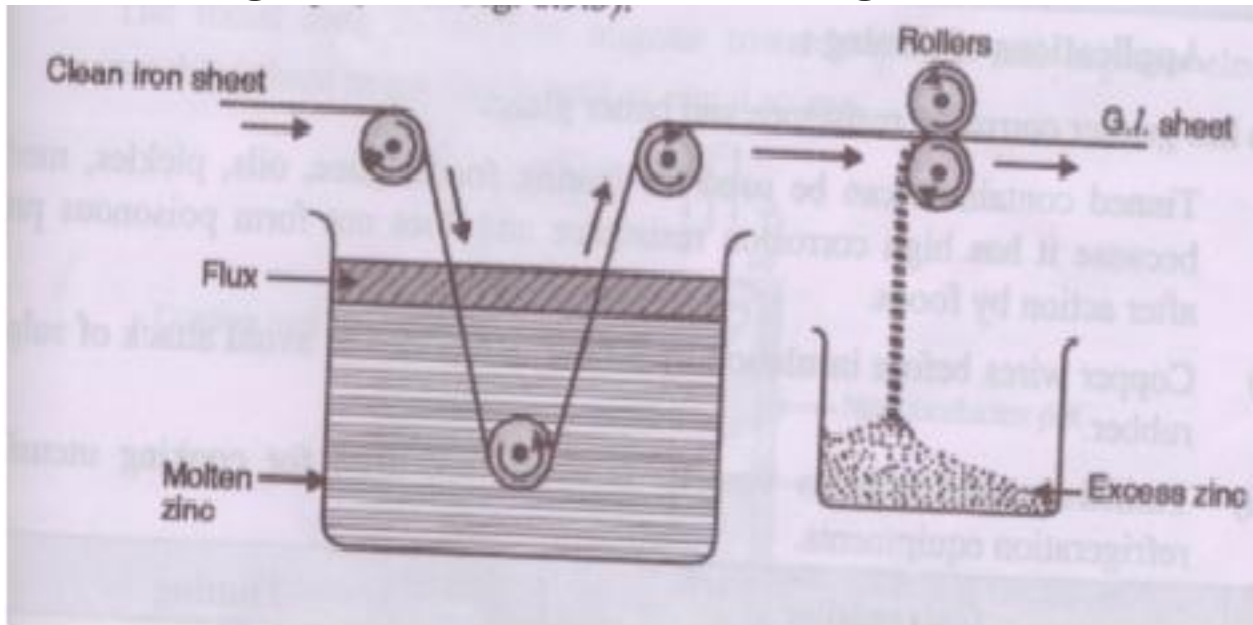
Used for producing a coating of low M.P metal.

Low melting metals – zinc- 419°C ; tin – 232°C

e.g. coating of Zn on iron – **Galvanizing**,
coating of Sn on iron – **tinning**,

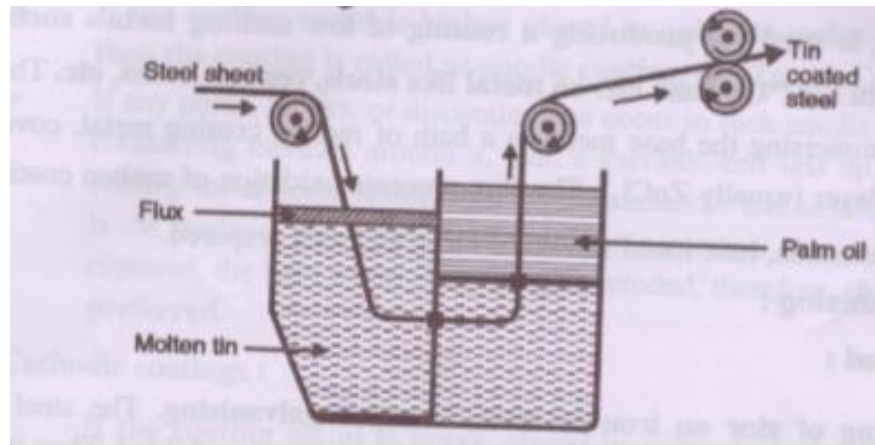
Galvanizing

- The process of coating iron or steel (base metal) with a thin coat of Zinc by hot dipping to prevent base metal corrosion is called galvanizing.
- **Process-** Zinc bath temperature $425 - 430^{\circ}\text{C}$
- **Use-** widely used for protection of iron – G.I. pipes, G.I. sheets buckets, tubes, screws, nails
- **Flux** – for adhesion and prevents the oxidation of molten coating metal after the coating .



Tinning

- Coating of tin on steel or iron is known as tinning.
- Tin bath temperature – 240°C
- Process -
- **Palm oil**- protect the hot tin coated surface from corrosion
- Uses- 1) tin coated containers can be used for storing food, ghee, oil, pickles, medicines . 2) copper wires before insulation by rubber are tinned 3) tinned copper sheets are used for making cooking utensils and refrigeration equipments



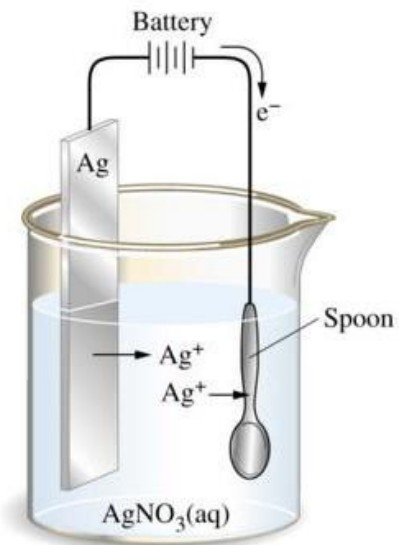
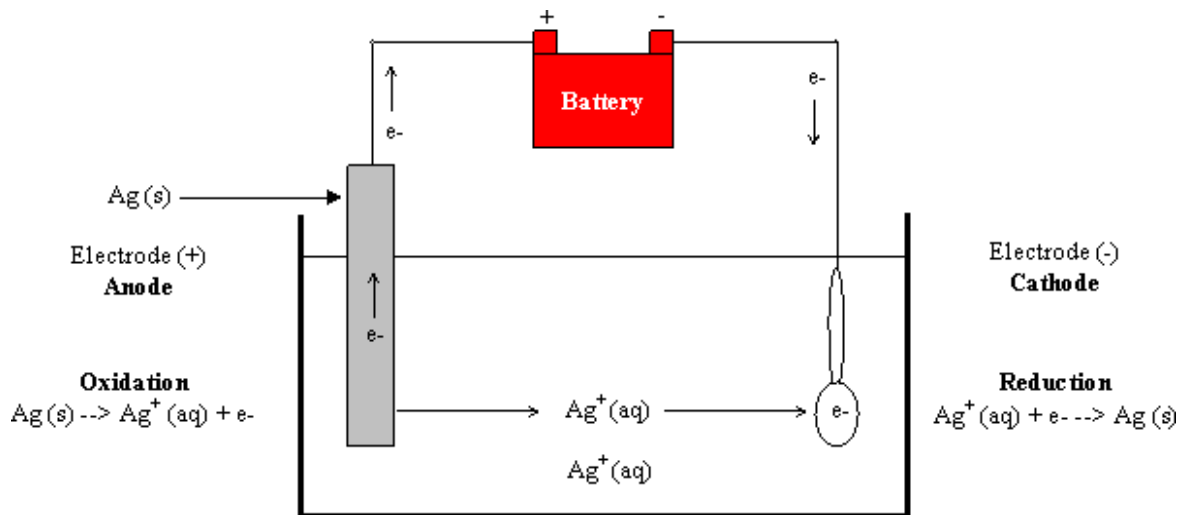
Difference between Galvanizing and tinning

Why galvanizing is better than tinning

Galvanizing	Tinning
1) Coating of Zn on Iron	1) Coating of Sn on iron
2) Anodic coating	2) Cathodic Coating
3) Reliable, Zn is higher placed in galvanic series	3) Less reliable, tin lower placed
4) Cheaper	4) Costly

Electroplating

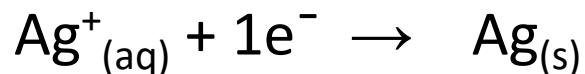
- Electroplating is method in which coating metal is coated on base metal by passing a direct current through the solution of an electrolyte containing the soluble salt of coating metal.
- Process- the article to be coated is cleaned first to remove oil, grease, oxides, scales etc.
- There is a non-conducting (plastic or concrete) tank containing solution of salt of the coating metal. The article is connected to the negative terminal of D.C. supply to make it cathode. Anode is the coating metal rod/ strip. After adjusting the suitable pH and density on the article surface, the electroplating starts .



- The anode reaction:



- The cathode reaction:



After adjusting the suitable pH and current density on article surface, the electroplating begins

	Reaction at anode	Reaction at cathode	Electrolyte salt
Nickel Plating	$\text{Ni} \longrightarrow \text{Ni}^{+2} + 2e^-$	$\text{Ni}^{+2} + 2e^- \longrightarrow \text{Ni}$	$\text{NiCl}_2, \text{NiSO}_4$
Cu Plating	$\text{Cu} \longrightarrow \text{Cu}^{+2} + 2e^-$	$\text{Cu}^{+2} + 2e^- \longrightarrow \text{Cu}$	CuSO_4
Ag Plating	$\text{Ag} \longrightarrow \text{Ag}^+ + e^-$	$\text{Ag}^+ + e^- \longrightarrow \text{Ag}$	AgCN

Applications of electroplating –

- 1) For getting decorative surface e.g. gold, chromium, brass coating on steel are used in ornaments, wrist watches, belts, pens etc.
- 2) Used for protecting surface against corrosion e.g. chromium plating on steel.
- 3) Nonmetallics like wood, paper, glass, leather, resins can be electroplated for decoration, preservation of surface, for obtaining conducting surface.

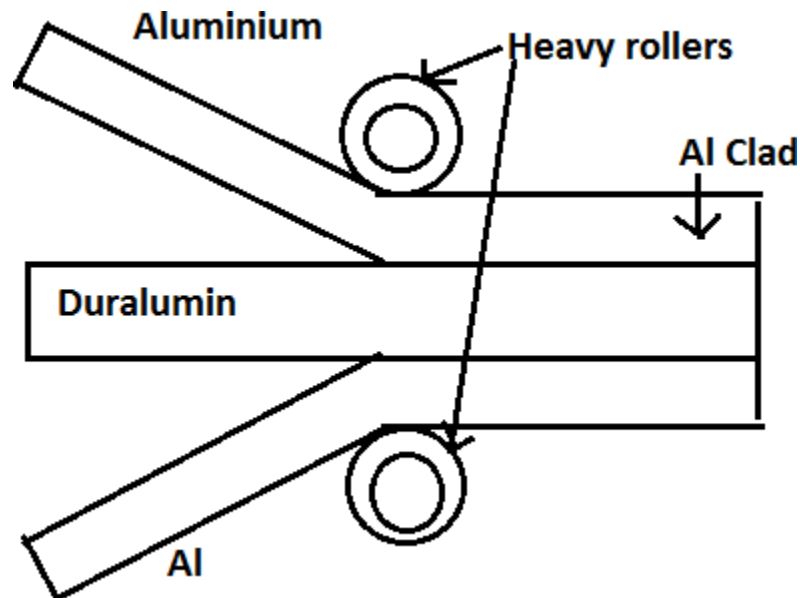
Advantages of electroplating –

- 1) Coating thickness can be controlled.
- 2) Coating of metal having high melting points like Cr, Ni, Ag, Au etc can be easily applied.
- 3) Process is not expensive.



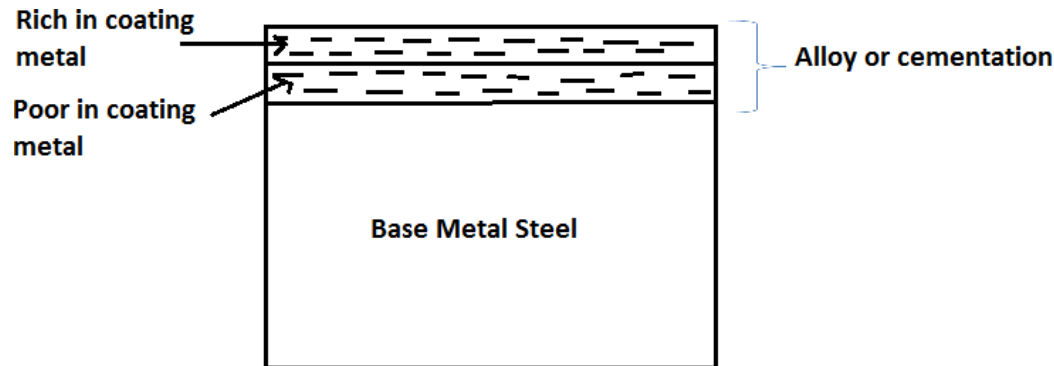
4) Metal cladding

- Cladding is done by sandwich base metal in between two layers of protecting metal, which are passed through rollers and bonded under the action of heat and pressure
- Clad metals- Ni, Pb, Ag, Cu, Cr, Al
- Nickel and Nickel alloy clad steel- handling chemicals
- Copper clad steel – used in electric industries
- Alclad sheets widely used in aircrafts industries



3) Cementation/ Diffusion coating

- In Cementation technique, there is a formation of a strong layer of alloy of coating metal and base metal
- Process- In this method the base metal article is packed in powder of coating metal and heated to a temperature just below the melting point of lower melting component. An alloy of the two metal is formed by diffusion of coating metal into the base metal.
- Coating metals like Zn, Cr, Al can be coated on iron.



Applications –

- 1) Coating metals , nuts, bolts, screws, spanner, screw drivers, turbine blades
- 2) Good abrasion resistance and strong

1) Sherardizing – Zn

2) Colorizing – Al

3) Chromizing – Cr

1) Sherardizing – Developed by Sherard Cowpercoles in 1890

1) Sherardizing - is the process of cementation, using zinc powder as coating metal.

In this process cleaned iron article is heated in closed drum with zinc powder at 350 -370°C for 2-3 hours. During this process, zinc get diffused into iron forming Zn-Fe alloy on the iron article surface. Dimension of article will not change after coating. Process is used for coating small articles like bolts, screws, nuts, washers, valves etc.

2) Colorizing – Colorizing is carried out by heating metal objects with roughened surface in a tightly packed drum with a mixture of aluminium powder and aluminium oxide, together with traces of NH_4Cl flux.

Method is carried out at $840\text{--}930^\circ\text{C}$ for 4 to 6 hrs.

Applicable for coating furnace parts, valves etc where resistance to oxidation and resistance to attack by sulphur at high temperature are required

3) Chromizing- process is carried out by heating the base metal with mixture of chromium powder and alumina at 1300°C for 3-4 hours.

this method is mostly applied for the protection of gas turbine blades.